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B.Tech. Sem I — Engineering Mathematics-I (BAS103 / KAS103T)

COMPREHENSIVE UNIT-WISE QUESTION BANK (PYQs 2020–2026)

UNIT-1: MATRICES — CONTACT HOURS: 8

Syllabus: *Elementary transformations, Inverse of a matrix, Rank of matrix, Solution of system of linear equations, Characteristic equation, Cayley-Hamilton Theorem and its application, Linear Dependence and Independence of vectors, Eigen values and Eigen vectors, Complex Matrices, Hermitian, Skew-Hermitian and Unitary Matrices, Applications to Engineering problems.*

SHORT ANSWER QUESTIONS (2 MARKS)

Q.NO.	QUESTION STATEMENT	YEAR & MARKS
1	State Rank-Nullity Theorem.	2020-21 | 2 M
2	Prove that the matrix $\frac{1}{\sqrt{3}}\begin{pmatrix} 1 & 1+i \\ 1-i & -1 \end{pmatrix}$ is unitary.	2020-21 | 2 M
3	If the matrix $A = \begin{pmatrix} -1 & 2 & 3 \\ 0 & 3 & 5 \\ 0 & 0 & -2 \end{pmatrix}$, then find the eigen values of $A^3 + 5A + 8I$.	2021-22 | 2 M
4	Reduce the matrix $\begin{pmatrix} 1 & 1 & 1 \\ 3 & 1 & 1 \end{pmatrix}$ into normal form and find its rank.	2021-22 | 2 M
5	If A is a Hermitian matrix, then show that iA is a Skew-Hermitian matrix.	2022-23 | 2 M
6	Find the eigen vector corresponding to the eigen value $\lambda = 6$ of the matrix $A = \begin{pmatrix} 4 & 2 \\ 2 & 4 \end{pmatrix}$.	2022-23 | 2 M
7	Find the product and sum of the eigen values for $A = \begin{pmatrix} 8 & -4 \\ 2 & 2 \end{pmatrix}$.	2023-24 | 2 M
8	Find the eigen values of the matrix $\begin{pmatrix} \cos\theta & -\sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$.	2024-25 | 2 M
9	In the matrix equation $\begin{pmatrix} 4 & 3 \\ 2 & x \end{pmatrix}\begin{pmatrix} y \\ -3 \end{pmatrix} = \begin{pmatrix} 5 \\ 1 \end{pmatrix}$, what are the values of x and y?	2025-26 | 2 M
10	Find the rank of the matrix $A = \begin{pmatrix} 2 & -4 & 6 \\ -1 & 2 & -3 \\ 3 & -6 & 9 \end{pmatrix}$.	2025-26 | 2 M

LONG ANSWER QUESTIONS (7 & 10 MARKS)

Q.NO.	QUESTION STATEMENT	YEAR & MARKS
11	Find the inverse of the matrix $A = \begin{pmatrix} 2 & 3 & 4 \\ 4 & 3 & 1 \\ 1 & 2 & 4 \end{pmatrix}$.	2020-21 | 10 M
12	Find non-singular matrices P and Q such that PAQ is in normal form, where $A = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & 1 & 1 \end{pmatrix}$.	2020-21 | 10 M
13	Find the eigen values and the corresponding eigen vectors of the matrix $A = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 3 & 0 \\ 1 & 0 & 2 \end{pmatrix}$.	2020-21 | 10 M
14	Find the characteristic equation of the matrix $A = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{pmatrix}$, prove that $A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I = \begin{pmatrix} 8 & 5 & 5 \\ 0 & 3 & 0 \\ 5 & 5 & 8 \end{pmatrix}$ and compute A^{-1}.	2021-22 | 10 M
15	Find the value of k for which the system of equations $(3k-8)x + 3y + 3z = 0$, $3x + (3k-8)y + 3z = 0$, $3x + 3y + (3k-8)z = 0$ has a non-trivial solution.	2021-22 | 10 M
16	Find the eigen values and eigen vectors of matrix $A = \begin{pmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ 3 & 3 & 4 \end{pmatrix}$.	2021-22 | 10 M
17	Verify Cayley-Hamilton theorem for the matrix $A = \begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix}$ and hence find its inverse.	2022-23 | 7 M
18	Test the consistency of the following system of equations and if consistent, solve them: $x + y + z = 6$, $x + 2y + 3z = 14$, $x + 4y + 7z = 30$.	2022-23 | 7 M
19	Find the eigen values and corresponding eigen vectors of the matrix $A = \begin{pmatrix} 2 & 1 & 1 \\ 2 & 3 & 4 \\ -1 & -1 & -2 \end{pmatrix}$.	2022-23 | 7 M
20	Solve the system of homogeneous equations: $x_1 + x_2 + x_3 + x_4 = 0$, $2x_1 + 3x_2 + 2x_3 + 4x_4 = 0$, $2x_1 + x_3 - x_4 = 0$.	2023-24 | 7 M
21	Determine eigen vectors for the matrix $A = \begin{pmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ -1 & 2 & 2 \end{pmatrix}$.	2023-24 | 7 M

Q.NO.	QUESTION STATEMENT	YEAR & MARKS
22	Determine A^{-1}, A^{-2} and A^{-3} if $A = \begin{pmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{pmatrix}$ using Cayley-Hamilton theorem.	2023-24 | 7 M
23	Find two non-singular matrices P and Q such that PAQ is in normal form, where $A = \begin{pmatrix} 1 & 3 & 6 & -1 \\ 1 & 4 & 5 & 1 \\ 1 & 5 & 4 & 3 \end{pmatrix}$.	2024-25 | 7 M
24	Find the eigen values and eigen vectors of the matrix $A = \begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix}$.	2024-25 | 7 M
25	Discuss for all values of K the system of equations $x + y + 4z = 6$, $x + 2y - 2z = 6$, $Kx + y + z = 6$ as regards existence and nature of solution.	2024-25 | 7 M
26	Determine the values of λ and μ such that the system: $2x - 5y + 2z = 8$, $2x + 4y + 6z = 5$, $x + 2y + \lambda z = \mu$ has (i) no solution, (ii) a unique solution, (iii) infinitely many solutions.	2025-26 | 7 M
27	Find the eigen values and eigen vectors of the matrix $A = \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$.	2025-26 | 7 M
28	If $X_1 = [3, 1, -4]^T$, $X_2 = [2, 2, -3]^T$, $X_3 = [0, -4, 1]^T$, show that the vectors X_1, X_2, X_3 are linearly dependent.	2025-26 | 7 M

UNIT-2: DIFFERENTIAL CALCULUS - I — CONTACT HOURS: 8

Syllabus: *Successive Differentiation (nth order derivatives), Leibnitz theorem, Curve tracing, Partial derivatives, Euler's Theorem for homogeneous functions, Total derivative, Change of variables.*

SHORT ANSWER QUESTIONS (2 MARKS)

Q.NO.	QUESTION STATEMENT	YEAR & MARKS
1	State Rolle's Theorem.	2020-21 | 2 M
2	Discuss all the symmetry of the curve $x^2y^2 = x^2 - a^2$.	2020-21 | 2 M
3	If $u = f(y-z, z-x, x-y)$, prove that $\partial u/\partial x + \partial u/\partial y + \partial u/\partial z = 0$.	2020-21 | 2 M
4	Find the envelope of the family of straight lines $y = mx + a/m$, where m is a parameter.	2021-22 | 2 M
5	Can mean value theorem be applied to $f(x) = \tan x$ in $[0, \pi]$?	2021-22 | 2 M
6	State Euler's Theorem on homogeneous functions.	2021-22 | 2 M
7	If $y = \cos^{-1}x$, prove that $(1-x^2)y_2 - xy_1 = 0$.	2022-23 | 2 M

Q.NO.	QUESTION STATEMENT	YEAR & MARKS
8	If $u = \sin^{-1}((x^3+y^3)/(\sqrt{x} + \sqrt{y}))$, show that $x(\partial u/\partial x) + y(\partial u/\partial y) = (5/2) \tan u$.	2022-23 | 2 M
9	Find all symmetry in the curve $y^2(a^2+x^2) = x^2(a^2-x^2)$.	2023-24 | 2 M
10	If $u = (x^2+y^2)/(x+y)$, find the value of $x(\partial u/\partial x) + y(\partial u/\partial y)$.	2024-25 | 2 M
11	What is the difference between total derivatives and partial derivatives?	2024-25 | 2 M
12	If $y = x^3 \log_e x$, prove that $d^4y/dx^4 = 6/x$.	2025-26 | 2 M
13	If $u(x,y) = (\sqrt{x} + \sqrt{y})^5$, then find the degree of the function u.	2025-26 | 2 M

LONG ANSWER QUESTIONS (7 & 10 MARKS)

Q.NO.	QUESTION STATEMENT	YEAR & MARKS
14	If $y = e^{\tan^{-1}x}$, prove that $(1+x^2)y_{n+2} + [(2n+2)x - 1]y_{n+1} + n(n+1)y_n = 0$.	2020-21 | 10 M
15	Find the envelope of the family of lines $x/a + y/b = 1$, where a and b are connected by the relation $a^n + b^n = c^n$.	2020-21 | 10 M
16	If $y = \sin(m \sin^{-1}x)$, find the value of y_n at $x = 0$.	2020-21 | 10 M
17	If $u = \sec^{-1}((x^3-y^3)/(x+y))$, prove that $x(\partial u/\partial x) + y(\partial u/\partial y) = 2 \cot u$. Also evaluate $x^2(\partial^2u/\partial x^2) + 2xy(\partial^2u/\partial x\partial y) + y^2(\partial^2u/\partial y^2)$.	2020-21 | 10 M
18	State Rolle's theorem and verify Rolle's theorem for the function $f(x) = (\sin x)/e^x$ in $[0, \pi]$.	2021-22 | 10 M
19	If $f(x) = x / (1 + e^{1/x})$ for $x \neq 0$ and $f(0) = 0$, show that the function is continuous but not differentiable at $x = 0$.	2021-22 | 10 M
20	If $y = (x + \sqrt{(1+x^2)})^m$, find $y_n(0)$.	2021-22 | 10 M
21	If $y\sqrt{(x^2-1)} = \log_e(x + \sqrt{(x^2-1)})$, prove that $(x^2-1)y_{n+1} + (2n+1)xy_n + n^2y_{n-1} = 0$.	2022-23 | 7 M
22	Trace the curve $x^2y^2 = (a^2+y^2)(a^2-y^2)$ in xy-plane, where a is constant.	2022-23 | 7 M
23	If $u = (x^2y^2)/(x^2+y^2) + \cos(xy/(x^2+y^2))$, prove that $x^2(\partial^2u/\partial x^2) + 2xy(\partial^2u/\partial x\partial y) + y^2(\partial^2u/\partial y^2) = 2(x^2y^2)/(x^2+y^2)$.	2022-23 | 7 M
24	If $u = y^2e^{y/x} + x^2\tan^{-1}(x/y)$, show that: (i) $x(\partial u/\partial x) + y(\partial u/\partial y) = 2u$ (ii) $x^2(\partial^2u/\partial x^2) + 2xy(\partial^2u/\partial x\partial y) + y^2(\partial^2u/\partial y^2) = 2u$.	2023-24 | 7 M
25	If $y = \cos(m \sin^{-1}x)$, prove that $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} + (m^2-n^2)y_n = 0$. Also find $(y_n)_0$.	2023-24 | 7 M
26	If $z = f(x,y)$ where $x = e^u + e^{-v}$ and $y = e^{-u} - e^v$, show that $\partial z/\partial u - \partial z/\partial v = x(\partial z/\partial x) - y(\partial z/\partial y)$.	2023-24 | 7 M
27	Find the nth derivative of $\tan^{-1}(x/a)$.	2024-25 | 7 M
28	Trace the curve $y^2(a+x) = x^2(3a-x)$.	2024-25 | 7 M

Q.NO.	QUESTION STATEMENT	YEAR & MARKS
29	If u = f(r), where $r^2 = x^2+y^2$, prove that $\partial^2u/\partial x^2 + \partial^2u/\partial y^2 = f''(r) + (1/r)f'(r)$.	2024-25 | 7 M
30	Find the nth derivative of $(2x+1) / [(2x-1)(2x+3)]$.	2025-26 | 7 M
31	If $z = x^4y^2\sin^{-1}(x/y) + \log x - \log y$, show that $x(\partial z/\partial x) + y(\partial z/\partial y) = 6x^4y^2\sin^{-1}(x/y)$.	2025-26 | 7 M
32	Trace the curve $y^2(a+x) = x^2(3a-x)$.	2025-26 | 7 M

UNIT-3: DIFFERENTIAL CALCULUS - II — CONTACT HOURS: 8

Syllabus: *Expansion of functions by Taylor's and Maclaurin's theorems for functions of one and two variables, Maxima and Minima of functions of several variables, Lagrange's method of multipliers, Jacobians, Approximation of errors.*

SHORT ANSWER QUESTIONS (2 MARKS)

Q.NO.	QUESTION STATEMENT	YEAR & MARKS
1	If $x = e^v \sec u$, $y = e^v \tan u$, evaluate $\partial(x,y)/\partial(u,v)$.	2020-21 | 2 M
2	Find the critical points of the function $f(x,y) = x^3 + y^3 - 3axy$.	2021-22 | 2 M
3	Find the percentage error in measuring the volume of a rectangular box when an error of 1% is made in measuring each side.	2022-23 | 2 M
4	Calculate the percentage error in R if E = RI and possible errors in E and I are 30% and 20% respectively.	2023-24 | 2 M
5	What are the applications of Jacobians?	2024-25 | 2 M
6	If $x = r\cos\theta$, $y = r\sin\theta$, find $\partial(r,\theta)/\partial(x,y)$.	2025-26 | 2 M

LONG ANSWER QUESTIONS (7 & 10 MARKS)

Q.NO.	QUESTION STATEMENT	YEAR & MARKS
7	If $u^3 + v + w = x + y^2 + z^2$, $u + v^3 + w^2 = x^2 + y + z^2$, $u + v + w^3 = x^2 + y^2 + z$, show that: $\partial(u,v,w)/\partial(x,y,z) = [1 - 4xy(xy+yz+zx) + 16xyz] / [2 - 3(u^2+v^2+w^2) + 27u^2v^2w^2]$.	2020-21 | 10 M
8	Divide 24 into three parts such that continued product of first, square of second and cube of third is a maximum.	2020-21 | 10 M
9	If u, v, w are roots of cubic equation $(\lambda-x)^3 + (\lambda-y)^3 + (\lambda-z)^3 = 0$ in λ, find $\partial(u,v,w)/\partial(x,y,z)$.	2021-22 | 10 M
10	Expand x^y in powers of (x−1) and (y−1) up to third-degree terms and hence evaluate $(1.1)^{1.02}$.	2021-22 | 10 M
11	A rectangular box open at the top has a capacity of 32 c.c. Find dimensions of the box such that least material is required for its construction.	2021-22 | 10 M
12	Expand $f(x,y) = y^x$ about (1,1) up to second degree terms and hence evaluate $(1.02)^{1.03}$.	2022-23 | 7 M

Q.NO.	QUESTION STATEMENT	YEAR & MARKS
13	Find Jacobian of functions $y_1 = (x_1-x_2)(x_2+x_3)$, $y_2 = (x_1+x_2)(x_2-x_3)$, $y_3 = x_2(x_1-x_3)$ and show they are functionally dependent. Find relation between them.	2022-23 | 7 M
14	A rectangular box, open at the top, has a capacity of 32 cu ft. Determine using Lagrange's method of multipliers the dimensions for least material.	2022-23 | 7 M
15	Expand $f(x,y) = e^x \cos y$ about point $(1, \pi/4)$ by Taylor's series.	2023-24 | 7 M
16	If $u^3 + v^3 = x + y$ and $u^2 + v^2 = x^3 + y^3$, show that $\partial(u,v)/\partial(x,y) = (y^2 - x^2) / [2uv(u-v)]$.	2023-24 | 7 M
17	Pressure P at any point (x,y,z) in space is $P = 400xyz^2$. Find the highest pressure on unit sphere $x^2+y^2+z^2 = 1$ using Lagrange's method.	2023-24 | 7 M
18	Find volume of largest rectangular parallelepiped inscribed in ellipsoid $x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$.	2024-25 | 7 M
19	If $u = xyz$, $v = x^2+y^2+z^2$ and $w = x+y+z$, find Jacobian $\partial(x,y,z)/\partial(u,v,w)$.	2024-25 | 7 M
20	Find maxima and minima of function $f(x,y) = \sin x + \sin y + \sin(x+y)$.	2024-25 | 7 M
21	Expand $x^2y + 3y - 2$ in powers of $(x-1)$ and $(y+2)$ up to 3rd degree using Taylor's series.	2025-26 | 7 M
22	Find minimum distance from point (1,2,0) to cone $z^2 = x^2 + y^2$.	2025-26 | 7 M
23	Find approximate value of $[(0.98)^2 + (2.01)^2 + (1.94)^2]^{1/2}$.	2025-26 | 7 M

UNIT-4: MULTIPLE INTEGRATION — CONTACT HOURS: 8

Syllabus: *Double integral, Triple integral, Change of order of integration, Change of variables, Beta and Gamma function and their properties, Dirichlet's integral and its applications to area and volume, Liouville's extensions of Dirichlet's integral.*

SHORT ANSWER QUESTIONS (2 MARKS)

Q.NO.	QUESTION STATEMENT	YEAR & MARKS
1	Evaluate $\int_0^1 \int_0^{x^2} e^{y/x}\, dy\, dx$.	2020-21 | 2 M
2	Calculate volume of solid bounded by surface x+y+z=1 and x=0, y=0, z=0.	2020-21 | 2 M
3	Find area bounded by curves $y^2 = x$ and $x^2 = y$.	2021-22 | 2 M
4	Find value of $\int_0^2 \int_0^2 \int_0^2 dx\, dy\, dz$.	2021-22 | 2 M
5	Evaluate $\iint y\, dx\, dy$ over plane bounded by line $y = x$ and parabola $y = 4x - x^2$.	2022-23 | 2 M
6	Determine the value of $\Gamma(1/4)\, \Gamma(3/4)$.	2023-24 | 2 M
7	Prove that $B(p,q) = B(p+1,q) + B(p,q+1)$.	2023-24 | 2 M
8	Write the statement of Liouville's Theorem.	2024-25 | 2 M
9	Evaluate $\int_1^2 \int_1^3 x^2 y^2\, dx\, dy$.	2024-25 | 2 M
10	Evaluate integral using Beta-Gamma function: $\int_0^2 x\,(8 - x^3)^{1/3}\, dx$.	2025-26 | 2 M

LONG ANSWER QUESTIONS (7 & 10 MARKS)

Q.NO.	QUESTION STATEMENT	YEAR & MARKS
11	Evaluate by changing variables: $\iint_R (x+y)^2$ dx dy where R is region bounded by parallelogram x+y=0, x+y=2, 3x−2y=0 and 3x−2y=3.	**2020-21 | 10 M**
12	Evaluate by changing order of integration: $\int_0^\infty \int_x^\infty (e^{-y} / y)$ dy dx.	**2020-21 | 10 M**
13	Triangular thin plate with vertices (0,0), (2,0) and (2,4) has density ρ = 1 + x + y. Find: (i) Mass of plate, (ii) Position of its centre of gravity G.	**2020-21 | 10 M**
14	Find volume bounded by cylinder $x^2 + y^2 = 4$ and planes y+z = 4 and z = 0.	**2021-22 | 10 M**
15	Change order of integration in $I = \int_0^1 \int_{x^2}^{2-x} xy$ dy dx and hence evaluate the same.	**2021-22 | 10 M**
16	Find position of C.G. of semicircular lamina of radius a if density varies as square of distance from diameter.	**2021-22 | 10 M**
17	Evaluate double integral $\int_0^a \int_{\sqrt{(ax)}}^a [y^2 / \sqrt{(y^4 - a^2x^2)}]$ dy dx by changing order of integration.	**2022-23 | 7 M**
18	Evaluate $\iiint_R (x - 2y + z)$ dz dy dx, where R is region determined by $0 \le x \le 1$, $0 \le y \le x^2$, $0 \le z \le x+y$.	**2022-23 | 7 M**
19	Use Dirichlet's integral to evaluate ∭ xyz dx dy dz throughout volume bounded by x=0, y=0, z=0 and x+y+z=1.	**2022-23 | 7 M**
20	Evaluate integral $\iint_D (y-x)$ dx dy by changing variables, where D is region bounded by y−x=−3, y−x=1, y+x/3=7/3 and y+x=5.	**2023-24 | 7 M**
21	Find volume of solid bounded by coordinate planes and surface $(x/a)^{1/2} + (y/b)^{1/2} + (z/c)^{1/2} = 1$.	**2023-24 | 7 M**
22	Prove that B(m,n) = Γ(m)Γ(n) / Γ(m+n).	**2023-24 | 7 M**
23	Apply Dirichlet's theorem to evaluate ∭ xyz dx dy dz throughout ellipsoid $x^2/a^2 + y^2/b^2 + z^2/c^2 \le 1$.	**2024-25 | 7 M**
24	Find area inside circle r = 2a cos θ and outside circle r = a.	**2024-25 | 7 M**
25	Change order of integration and evaluate $\int_0^{2a} \int_{x^2/(4a)}^{3a-x} (x^2+y^2)$ dy dx.	**2024-25 | 7 M**
26	Change order of integration and evaluate $\int_0^a \int_{\sqrt{(ax)}}^a [y^2 / \sqrt{(y^4 - a^2x^2)}]$ dy dx.	**2025-26 | 7 M**
27	Prove area in positive quadrant bounded by $y^2=4ax$, $y^2=4bx$, $xy=c^2$ and $xy=d^2$ is $(1/3)(d^2-c^2)\log(b/a)$, where d>c, b>a.	**2025-26 | 7 M**
28	Find volume of solid bounded by coordinate planes and surface $\sqrt{(x/a)} + \sqrt{(y/b)} + \sqrt{(z/c)} = 1$.	**2025-26 | 7 M**

UNIT-5: VECTOR CALCULUS — CONTACT HOURS: 8

Syllabus: *Vector differentiation: Gradient, Curl and Divergence and their Physical interpretation, Directional derivatives. Vector Integration: Line integral, Surface integral, Volume integral, Gauss's Divergence theorem, Green's theorem and Stoke's theorem (without proof) and their applications.*

SHORT ANSWER QUESTIONS (2 MARKS)

Q.NO.	QUESTION STATEMENT	YEAR & MARKS
1	Show that vector field $\vec{V} = (x+3y)\hat{i} + (y-3z)\hat{j} + (x-2z)\hat{k}$ is solenoidal.	2020-21 | 2 M
2	State Green's theorem.	2020-21 | 2 M
3	Find a unit normal vector to surface $z^2 = x^2 + y^2$ at point (1, 0, −1).	2021-22 | 2 M
4	State Stoke's Theorem.	2021-22 | 2 M
5	Find curl of vector field $\vec{F} = (x^2+xy^2)\hat{i} + (y^2+x^2y)\hat{j}$.	2022-23 | 2 M
6	Prove that vector $\vec{A} = (6xy+z^3)\hat{i} + (3x^2-z)\hat{j} + (3xz^2-y)\hat{k}$ is irrotational.	2023-24 | 2 M
7	Find a unit normal vector to surface $xy^3z^2 = 4$ at point (−1,−1, 2).	2023-24 | 2 M
8	Prove that curl $\vec{r} = 0$, where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$.	2024-25 | 2 M
9	Show that gradient field describing a motion is irrotational.	2025-26 | 2 M

LONG ANSWER QUESTIONS (7 & 10 MARKS)

Q.NO.	QUESTION STATEMENT	YEAR & MARKS
10	Use divergence theorem to evaluate surface integral $\iint_S (x\,dy\,dz + y\,dz\,dx + z\,dx\,dy)$ where S is portion of plane x+2y+3z=6 in first octant.	2020-21 | 10 M
11	A fluid motion is given by $\vec{v} = (y\sin z - \sin x)\hat{i} + (x\sin z + 2yz)\hat{j} + (xy\cos z + y^2)\hat{k}$. Is motion irrotational? If so, find velocity potential.	2020-21 | 10 M
12	Verify Stoke's theorem for function $\vec{F} = x^2\hat{i} + xy\hat{j}$ integrated round square whose sides are x=0, y=0, x=a, y=a in plane z=0.	2020-21 | 10 M
13	Apply Green's theorem to evaluate $\oint_C [(2x^2-y^2)dx + (x^2+y^2)dy]$ where C is boundary enclosed by x-axis and upper half of circle $x^2+y^2=a^2$.	2021-22 | 10 M
14	Find directional derivative of $\nabla(\nabla f)$ at (1,−2,1) in direction of normal to surface $xy^2z = 3x + z^2$ where $f = 2x^3y^2z^4$.	2021-22 | 10 M
15	Find constants a, b, c so that $\vec{F} = (x+2y+az)\hat{i} + (bx-3y-z)\hat{j} + (4x+cy+2z)\hat{k}$ is irrotational and find function ϕ such that $\vec{F} = \nabla\phi$.	2021-22 | 10 M
16	Find directional derivative of scalar function f(x,y,z) = xyz at point P(1,1,3) in direction of outward normal to sphere $x^2+y^2+z^2=11$ through P.	2022-23 | 7 M
17	Apply Gauss divergence theorem to evaluate $\iint_S \vec{F}\cdot\hat{n}\,dS$, where $\vec{F} = 4x\hat{i} - 2y^2\hat{j} + z^2\hat{k}$ and S is cylinder $x^2+y^2=4$, z=0, z=3.	2022-23 | 7 M

Q.NO.	QUESTION STATEMENT	YEAR & MARKS
18	Evaluate $\oint_C \vec{F} \cdot d\vec{r}$ by Stoke's theorem, where $\vec{F} = y^2\hat{i} + x^2\hat{j} - (x+z)\hat{k}$ and C is triangle with vertices (0,0,0), (1,0,0), (1,1,0).	**2022-23 | 7 M**
19	Find directional derivative of $f(x,y,z) = e^{2x} \cos yz$ at (0,0,0) in direction of tangent to curve $x = a \sin\theta$, $y = a \cos\theta$, $z = a\theta$ at $\theta = \pi/4$.	**2023-24 | 7 M**
20	Applying Gauss Divergence theorem, evaluate $\iint_S [e^x\, dy\, dz - y\, e^x\, dz\, dx + 3z\, dx\, dy]$, where S is surface of cylinder $x^2+y^2=c^2$, $0 \le z \le h$.	**2023-24 | 7 M**
21	Prove that $\nabla^2 r^n = n(n+1)\, r^{n-2}$, where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and show that $\nabla^2(1/r) = 0$.	**2023-24 | 7 M**
22	Show that vector field $f(r)\vec{r}$ is irrotational, where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$.	**2024-25 | 7 M**
23	Show that $\text{div}(\text{grad}\, r^n) = n(n+1)\, r^{n-2}$, where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$.	**2024-25 | 7 M**
24	Verify Stokes theorem for $\vec{F} = (x^2+y^2)\hat{i} - 2xy\hat{j}$ taken round rectangle bounded by x=0, x=a, y=0, y=b.	**2024-25 | 7 M**
25	Find directional derivative of divergence of $\vec{F}(x,y,z) = xy\hat{i} + xy^2\hat{j} + z^2\hat{k}$ at (2,1,2) in direction of outer normal to sphere $x^2+y^2+z^2=9$.	**2025-26 | 7 M**
26	Apply Green's theorem to evaluate $\int_C [(y - \sin x)dx + \cos x\, dy]$ where C is triangle formed by y=0, $x=\pi/2$, $y = (2/\pi)x$.	**2025-26 | 7 M**
27	Apply Gauss's divergence theorem to evaluate $\iint_S \vec{F} \cdot \hat{n}\, dS$, where $\vec{F} = (2x^2-3z)\hat{i} - 2xy\hat{j} - 4x\hat{k}$ and S is bounded by planes x=0, y=0, z=0 and 2x+2y+z=4.	**2025-26 | 7 M**